

# Veronica Pimenova

Information Systems & Human-Computer Interaction (HCI)  
Carnegie Mellon University

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## Education

**Carnegie Mellon University, Pittsburgh, PA** **August 2022 - May 2025**

*Bachelor of Science in Information Systems & minor in Human-Computer Interaction*

**Advised by Andrew Begel, GPA: 3.7/4.0**

Dean's List (High Honors)

**Relevant Courses:** Imperative Computation, Reasoning with Data, Human-Centered Software, Database Design & Development, Empirical Research Methods, Application Design & Development

**Languages:** Python, Java, JavaScript, SQL, TypeScript, HTML, CSS

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## Research Experience

**Principal Investigator, CMU Software & Societal Systems** **Sept 2022 – Present**

PI: Veronica Pimenova (Advised by Andrew Begel). Conducted a research study with 20+ participants, encompassing participant recruitment, experimental design, and statistical data analysis. Developed and tested an AI-driven accessible learning platform tailored for neurodiverse populations, specifically targeting ADHD and autism. Implemented machine learning models for action segmentation, leveraging temporal and linguistic cues to detect key instructional steps in Scratch-based pedagogical videos, enabling adaptive video content delivery.

**Research Assistant, CMU Robotics Institute** **May 2024 – Oct 2024**

PI: Andrew Hundt. Conducted research on Autism Spectrum Disorders, emphasizing data-driven insights through literature reviews and Python-based analysis. Collaborated with interdisciplinary teams to refine experimental methodologies and optimize data processing pipelines. Designed and implemented Python tools for automated data analysis and visualization, streamlining the interpretation of complex research findings.

**Student Researcher, CMU Decision Sciences** **Sept 2023 – Dec 2023**

PI: Erin Carbone. Led a team of six researchers to investigate the correlation between grade fairness and student quality of life. Organized and facilitated bi-weekly meetings to coordinate tasks, conduct comprehensive literature reviews, and synthesize findings. Designed and presented a research poster summarizing key insights to an audience of 100+ students, showcasing data-driven conclusions and their implications for educational policy.

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## Publications

**Pimenova, V., Lee, C., Bhakdibhumi, B., Chu, S., Begel, A.** "Enhancing Learning Platforms for Individuals with ADHD." Submitted to the International Computing Education Research (ICER) Conference, under review.

Hundt, A., Ohlson, G., Wolfert, P., Miranda, L., Zhu, S., **Pimenova, V.**, Winkle, K., & Lee, C. "Love, Joy, and Autism Robots: A Metareview." Submitted to ACM Transactions on Human-Robot Interaction (THRI), under review.

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## Conference Presentations

- Pimenova, V.**, Bhakdibhumi, A., & Begel, A. (2023, 2024). "Neuroinclusive Documentation." Presented at Carnegie Mellon University Meeting of the Minds (MoM).
- Pimenova, V.**, Ranaweera, L., & Walker, D. (2022). "Computer Vision in Sports Motion Analysis." Presented at the Pittsburgh Regional Science & Engineering Fair (PRSEF).
- Pimenova, V.**, & Griest, P. (2020). "Synthesis of Spiky Nanoparticles for Safer Paints." Presented at PRSEF.
- Pimenova, V.**, & Griest, P. (2019). "Optimized Fluorescent Paint Design with Six Sigma." Presented at PRSEF.
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## Invited Talks

- Pimenova, V.** (2024, October). Workshop presentation in collaboration with CyLab [Workshop presentation]. OurCS, Carnegie Mellon University.
- Pimenova, V.** (2024, September). Guest lecture on accessibility research to 200+ attendees during CMU's Family Weekend [Oral presentation]. Carnegie Mellon University.
- Pimenova, V.** (2024, September). Guest lecture on accessibility in software design [Oral presentation]. Carnegie Mellon University, 67-200 Course.
- Pimenova, V.** (2023, October). Panelist for software internships talk [Panel discussion]. Carnegie Mellon University.
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## Professional Experience

- Incoming Associate Product Manager Intern, Pegasystems** **January 2025 - May 2025**  
Location: Remote  
Will collaborate with the Customer Decision Hub team to design and prioritize data-driven product features that optimize customer engagement and enhance decision-making workflows. Will focus on translating user insights into actionable feature roadmaps and aligning technical implementations with business objectives to ensure sustainable practices.
- Software Engineering Intern, Forevergreen** **May 2024 - Aug 2024**  
Location: Pittsburgh, PA  
Developed Python backend services for a carbon calculator application to support environmental sustainability efforts, enabling individuals and organizations to track and reduce their carbon footprints. Optimized data storage by 30% to handle large-scale environmental datasets efficiently. Designed and implemented REST APIs to ensure seamless communication between systems and integrated interactive front-end components in TypeScript to enhance user engagement with sustainability insights.
- Machine Learning Intern, Velo AI** **May 2023 - Aug 2023**  
Location: Pittsburgh, PA  
Processed and analyzed large-scale video datasets to enhance social media campaign performance, resulting in a twofold increase in product visibility. Designed and deployed machine learning models for object detection and animation overlays, leveraging state-of-the-art algorithms. Conducted field research and iterative demo testing in the Pittsburgh area to refine algorithmic accuracy and validate real-world application of features.

## **Software Engineering Intern, Argo AI**

**May 2022 - Aug 2022**

Location: Pittsburgh, PA

Developed and trained a lane-blocking classifier using random forest algorithms on large-scale real-world video datasets to improve autonomous vehicle decision-making. Enhanced Python scripts for data preprocessing, achieving significant improvements in data filtering efficiency and model readiness. Collaborated with the perception team to integrate and deploy machine learning models into a production environment, ensuring seamless operation within autonomous driving systems.

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## **Fellowships and Grants**

### **\$20,000 MSA Deike-Ryan Scholarship Program**

Issued by Mine Safety Appliances (MSA) Inc., awarded for outstanding academic performance, leadership, and volunteer work.

### **\$10,000 Raytheon Technologies Underrepresented Minorities Award**

Issued by International Information System Security Certification Consortium (ISC<sup>2</sup>), recognizing achievements in accessibility research, academic performance, and volunteer work in promoting underrepresented minorities in tech.

### **\$5,000 ISC<sup>2</sup> Undergraduate Scholarship**

Awarded by International Information System Security Certification Consortium for excellence in accessibility research, academic performance, and leadership in promoting underrepresented minorities in tech.

### **\$4,500 Boeing Leadership Award for Information Systems**

Issued by The Boeing Company for exemplary leadership and academic performance in the CMU Information Systems program.

### **\$4,500 Undergraduate Research Fellowship**

Awarded by Carnegie Mellon University, Office of Undergraduate Research and Scholar Development, for conducting advanced undergraduate research projects.

### **\$4,000 Octillo Women's Cybersecurity Award**

Recognized by International Information System Security Certification Consortium (ISC<sup>2</sup>) for contributions to the field of information, promoting diversity, and undergraduate research.

### **\$2,500 Cranberry Cup Scholarship**

Issued by Cranberry Cup Organization, supporting academic achievements and community service.

### **\$2,000 Bess Family Funding Recipient**

Awarded by Bess Family Fund for the development of machine learning models for autonomous software.

### **\$1,000 Tapia Conference Scholarship**

Sponsored by Carnegie Mellon Information Systems Inclusion Committee to attend the 2024 Tapia Conference.

### **\$400 XR Access Symposium Scholarship**

Provided by Cornell University and Meta Reality Labs for participation in discussions on accessibility in Augmented and Virtual reality (AR, VR).

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## Awards & Honors

### **Women and Information Technology Affiliate Award (3x)**

Issued by the National Center for Women and Information Technology (NCWIT) for promoting diversity in computing.

### **3rd Place in Computer Science, PRSEF**

Awarded at the Pittsburgh Regional Science & Engineering Fair, Spring 2022, for a distinguished computer science project utilizing OpenCV and Python to develop a system that identifies and mitigates bias in U.S. figure skating scoring processes.

### **PRSEF Corporate Sponsorship Awards (2x)**

Received at the Pittsburgh Regional Science & Engineering Fair for innovative computer science projects.

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## Teaching Experience

### **Head Teaching Assistant (Incoming), Carnegie Mellon University** **January 2025 - May 2025**

*Course: Information Systems Milieux (67-250), Dietrich College Information Systems*

Supervised by Dr. Jeria L. Quesenberry. Will oversee and lead recitations for a course covering database management, Python programming, and information systems concepts for a class of 100+ students. Responsibilities include coordinating curriculum development, ensuring technical accuracy in course content, and mentoring teaching assistants to deliver consistent and high-quality instruction.

### **Teaching Assistant, Carnegie Mellon University** **January 2024 - May 2024**

*Course: Information Systems Milieux (67-250), Dietrich College Information Systems*

Supervised by Dr. Jeria L. Quesenberry. Recitations focused on database management, Python programming, and information systems principles for a class of 100+ students. Provided individualized support to reinforce technical concepts such as SQL queries, Python-based data analysis, and systems design. Assisted in grading assignments, exams, and projects, ensuring fair and accurate assessment of technical proficiency. Organized review sessions to clarify course material and address student questions related to practical implementation.

### **Teaching Assistant, Carnegie Mellon University** **May 2024 - August 2024**

*Course: Undergraduate Research (99-370), Summer Undergraduate Research Apprenticeship (SURA)*

Supervised by Dr. Richelle Bernazzoli. Supported students in formulating research questions, identifying appropriate research methodologies, and developing skills to write research proposals. Facilitated weekly check-ins to track progress and address challenges, ensuring successful completion of individual research projects.

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## Conferences Attended

### **ACM SIGASSETS 2024, Virtual** **October 2024**

Served as a virtual Student Volunteer (SV) at the ACM Special Interest Group on Accessible Computing (SIGACCESS) Conference. Assisted with organizing sessions and supporting accessibility initiatives by making sure video submissions followed accessibility guidelines.

### **UIST 2024 Conference, Pittsburgh, PA** **October 2024**

Attended the ACM Symposium on User Interface Software and Technology as a sponsored participant for MIT Media Lab's workshop on soft wearables. Engaged in discussions on emerging trends in HCI and wearable technology, with a focus on accessibility.

**Tapia Conference 2024, San Diego, CA****September 2024**

Participated in the Richard Tapia Celebration of Diversity in Computing, engaging with talks on diversity, inclusion, and accessibility in computing. Presented research to attendees and networked with professionals and academics.

**XR Access Symposium 2024, New York, NY****June 2024**

Attended the XR Access Symposium, focusing on enhancing accessibility in virtual and augmented reality. Participated in workshops and discussions on inclusive design for immersive technologies. Presented to hundreds of in-person and remote attendees.

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**Volunteer Experience****Member, AccessComputing, University of Washington****September 2024 – Present**

Collaborating with a nationwide network to promote accessibility in computing education and research. Participating in workshops, discussions, and initiatives aimed at improving inclusion and accessibility for individuals with disabilities in computing fields.

**Notetaker, XR Access****June 2024**

Documented key discussions and presentations focused on improving accessibility in virtual and augmented reality technologies.

**Administrative Assistant, Academic Games League of America****April 2024**

Assisted in organizing national academic competitions, managing event logistics, and providing operational support for attendees and staff.

**Lead Mentor, Women in Information Systems, Carnegie Mellon University****Aug 2023 – May 2024**

Organized mentorship opportunities for female students in technology, leading workshops on career development, networking, and technical skill-building.

**DEI Chair, Delta Delta Delta Sorority, Carnegie Mellon University****Aug 2023 – Dec 2023**

Advocated for inclusivity by implementing Diversity, Equity, and Inclusion (DEI) strategies for 70+ members, coordinating events and fostering a supportive community.

**UI/UX Team Member, CMU Scotty Labs****Sept 2022 – May 2023**

Participated in design workshops and contributed to hackathon projects, including Tartanhacks, focusing on innovative user interface designs.

**Java & Python Instructor, Steel City Codes, Pittsburgh, PA****June 2021 – Aug 2021**

Taught Java programming through game design to middle and high school students from underrepresented backgrounds, fostering interest in technology.

**Lead Performer, Ronald McDonald House Charities****June 2021 – Aug 2021**

Performed figure skating musical and artistic showcases at fundraising and community events to support the organization's mission of providing housing for families in need.